

Deep Learning

Mock examination questions

Each question carries 25 marks and should take about 40 minutes, as will the deep learning question in the examination. Solutions begin on page 5.

Question 1

An agricultural robotics company is developing a machine that sprays herbicide on weeds alone, leaving the crop and the soil untouched. A camera beneath the machine photographs the ground, and for each photograph the machine needs a mask of the same size in which every pixel is marked as either weed or not weed. To train a network that produces these masks, agronomists have drawn masks by hand for 2,000 colour photographs of 128×128 pixels, all of which were taken in fields of lettuce on the company's own farm.

As a deep learning engineer you are asked to advise the company on various aspects of this task.

- (a) A member of the company's staff suggests flattening each photograph into a vector and using fully connected layers throughout, with 100 units in the first hidden layer. Calculate the number of weights in this first layer, and compare it with the number of weights in a 3×3 convolutional layer that produces 16 feature maps from the same photograph. Explain why convolutional layers are the more suitable choice for this task. **[4 Marks]**
- (b) Propose a convolutional encoder–decoder network that takes a photograph and produces its mask. Describe how the spatial size and the number of channels change through the network, how its output represents the mask, and which loss function you would use. **[6 Marks]**
- (c) The first network that the company trains passes information from its encoder to its decoder through the narrowest layer alone. Its masks show roughly where the weeds are, but their outlines are blurred and thin leaves are missed altogether. Explain why this happens, and propose a change to the network that would give sharper masks, explaining why it helps. **[5 Marks]**
- (d) A network with skip connections is trained on 1,600 of the photographs, and the remaining 400 are held out for validation. After several epochs the training loss continues to fall, whereas the validation loss begins to rise. Explain what this indicates, describe how data augmentation could be used to address it for these photographs, including one precaution that this task requires, and suggest one further remedy. **[5 Marks]**
- (e) The company intends to sell the machine to farms that grow other crops in other types of soil, where it will be used throughout the growing season. Assess whether the performance measured on the held-out photographs is a reliable guide to the performance that can be expected on these farms, and suggest how the company could check and improve that performance. **[5 Marks]**

Question 2

A games studio is developing a game in which players collect imaginary creatures, each drawn as a colour image of 64×64 pixels. Its artists have drawn 20,000 creatures by hand, and the studio is considering the use of a generative model to produce new and varied creatures in the same style.

As a deep learning engineer you are asked to advise the studio on various aspects of this task.

- (a) The studio first considers a generative adversarial network (GAN), which is trained with the objective

$$\min_G \max_D \mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)] + \mathbb{E}_{z \sim p_z} [\log (1 - D(G(z)))].$$

Explain what x , z , $G(z)$ and $D(x)$ represent, and explain the purpose of each of the two terms in the training of the two networks. Describe how the two networks are updated in each training iteration. **[7 Marks]**

- (b) A colleague suggests using either a diffusion model or a flow matching model instead. For one of these, outline how it would be trained on the images of creatures and how new images would then be generated. **[6 Marks]**

- (c) The GAN that the studio trains gives sharp images, but almost every one of them shows the same blue bird. The studio therefore wishes to compare the available types of model before choosing one for each of two uses. The first is a design tool for its artists, which may take a minute to produce an image but must give sharp images that cover the full variety of creatures in the training set. The second is a feature of the game itself, which must produce a new creature on a player's phone within a few milliseconds.

Compare GANs, variational autoencoders and diffusion (or flow matching) models in terms of the quality of their images, their coverage of the variety in the training data, and the time that they take to generate an image. Recommend a model for each of the two uses, justifying your choices. **[8 Marks]**

- (d) The studio would also like to choose the type of creature that is generated (bird, fish, insect or reptile), and to create new creatures that blend two of these types. Explain how the model from part (b) could be adapted to allow this, and comment on how reliable the blended creatures are likely to be. **[4 Marks]**

Question 3

A mobile phone manufacturer is considering a new feature for its on-screen keyboard, which would suggest how the message that a user is typing might continue. It plans to train a small Transformer to predict the next character of a message, using a collection of text messages donated by volunteers.

As a deep learning engineer you are asked to advise the manufacturer on various aspects of this task.

- (a) Describe how a passage of training text is converted into the inputs and targets used to train the model, illustrating your answer with the word `hello`, and state a suitable loss function. **[4 Marks]**
- (b) Explain how self-attention allows the prediction at one position to make use of the characters that precede it, referring in your answer to queries, keys and values. Explain also why a causal mask is needed during training, even though the whole of each message is available. **[7 Marks]**
- (c) To measure performance, an engineer forms windows of 256 consecutive characters starting at every position in the collection, and sets aside a randomly chosen 10% of these windows for validation. The validation loss is very low, yet the keyboard performs poorly when it is tried on new messages. Explain why the validation loss is misleading, and describe how the data should have been divided. **[4 Marks]**
- (d) Describe how the trained model would be used to generate a suggestion that is several words long, and explain how the sampling temperature affects the suggestions. **[4 Marks]**
- (e) The phone can store a model of at most one million parameters. Three models are trained on the same data and evaluated on a validation set that has been separated correctly. The results are shown below, where the loss is the mean negative log-likelihood per character.

Model	Parameters	Training loss	Validation loss
A: one block of width 64	99,328	1.80	1.83
B: four blocks of width 128	890,368	1.32	1.45
C: eight blocks of width 256	6,509,312	0.85	1.71

Evaluate the three models, recommend one of them for the keyboard, and propose two further changes that might improve the recommended model. **[6 Marks]**

Question 4

A mapping company stores many millions of aerial photographs, each a colour image of 256×256 pixels, and wishes to reduce the cost of storing them. Its engineers are considering replacing each photograph by a small neural network that maps the coordinates of a pixel to its colour, so that only the weights of the network need to be stored.

As a deep learning engineer you are asked to advise the company on various aspects of this task.

- (a) Outline how such a network would be fitted to a single photograph, and how the photograph would later be recovered from the stored weights. **[6 Marks]**
- (b) The proposed network has two inputs, two hidden layers of 64 units each and three outputs, and every layer has a bias. Calculate the number of parameters in the network, and compare it with the number of values needed to store the photograph directly. **[4 Marks]**
- (c) When ReLU activations are used, the recovered photographs show the large regions of colour correctly, but roads, hedges and buildings are blurred. Explain why this happens, and propose two changes to the network that could recover finer detail without increasing the number of stored values. **[6 Marks]**
- (d) The company measures quality using the peak signal-to-noise ratio, $\text{PSNR} = -10 \log_{10}(\text{MSE})$, where MSE is the mean squared error between the recovered and original photographs when intensities lie between 0 and 1, and it requires a PSNR of at least 30 dB. Calculate the PSNR of a recovered photograph whose MSE is 0.01, and state the MSE at which the requirement would be met. **[2 Marks]**
- (e) A manager observes that the network can be evaluated at any coordinate, and claims that rendering it at 1024×1024 pixels will reveal detail that was absent from the original photograph. Assess this claim, and describe how it could be tested. **[3 Marks]**
- (f) Fitting a network takes several thousand optimisation steps for each photograph. Explain which property of this approach makes it so expensive for a collection of millions of photographs, and propose a design in which a single trained network serves every photograph. **[4 Marks]**

Solutions

Question 1

(a) *Comprehension, Application*

Flattening the RGB photograph gives $3 \times 128 \times 128 = 49,152$ inputs, so 100 hidden units require $49,152 \times 100 = 4,915,200$ weights. [1 mark] A convolution with 16 output channels instead needs $3 \times 3 \times 3 \times 16 = 432$ weights. [1 mark]

Sharing each convolutional kernel across positions lets the network detect the same leaf feature wherever it appears, with far fewer weights to estimate from the limited training set. [1 mark] Local connections also build in the spatial relationships useful for finding plant outlines. Flattening preserves the pixel values, but fully connected layers do not encode this local structure in their connectivity. [1 mark]

Counting biases as well gives 4,915,300 and 448 parameters respectively. An explanation that convolutions can process different spatial sizes is also acceptable for the final mark.

(b) *Knowledge, Application, Synthesis*

Use an encoder with stride-two convolutions followed by ReLU, for example kernel size 4, stride 2 and padding 1. [1 mark] A suitable progression is $3 \times 128 \times 128 \rightarrow 32 \times 64 \times 64 \rightarrow 64 \times 32 \times 32 \rightarrow 128 \times 16 \times 16$, increasing channels as spatial size decreases. [1 mark]

A decoder can use stride-two transposed convolutions, or upsampling followed by ordinary convolutions. [1 mark] It doubles the spatial size at each stage while reducing the channel count, ending at $1 \times 128 \times 128$. [1 mark] Apply a sigmoid to interpret each output as the probability of weed, then threshold it to obtain the binary mask. [1 mark] Train with pixelwise binary cross-entropy against the hand-drawn mask, preferably passing raw logits to `BCEWithLogitsLoss` for numerical stability. [1 mark]

A two-channel softmax output with cross-entropy is equally suitable. Mean squared error against the binary mask is also creditable.

(c) *Analysis, Synthesis*

Downsampling reduces the number of spatial positions, making precise boundary locations harder to retain through the bottleneck. [1 mark] The decoder then has to reconstruct thin leaves and edges from coarse features, and uncertainty about their positions can produce blurred probabilities or missed structures. [1 mark]

Add encoder–decoder skip connections as in a U-Net. [1 mark] Pass features from an encoder stage to the decoder stage at the same resolution, concatenating them or adding them when channel counts match. [1 mark] This lets the decoder combine the finer spatial information from early layers with deeper features that identify weeds. [1 mark]

A less restrictive bottleneck is an acceptable alternative if the explanation connects it to better retention of spatial detail.

(d) *Comprehension, Application*

Falling training loss with rising validation loss indicates overfitting: further updates are improving the fit to these training photographs while worsening predictions on the held-out ones. [1 mark]

Data augmentation exposes the model to varied versions of its training images, encouraging it to retain useful patterns across those changes. [1 mark] Plausible rotations and flips vary plant orientation, while modest brightness changes imitate different lighting. [1 mark] Geometric transformations must also be applied to the masks so their labels remain aligned with the pixels. Brightness changes should affect the photograph only. [1 mark]

Early stopping can retain the checkpoint with the lowest validation loss. [1 mark] Weight decay, dropout or additional labelled photographs are acceptable further remedies. Avoiding unrealistic augmentations is an alternative valid precaution.

(e) Analysis, Evaluation

The held-out photographs come from the same lettuce fields as the training set, so they mainly assess performance under those conditions. [1 mark] Other farms may differ in crop, soil, weed appearance, growth stage and lighting. [1 mark] These distribution changes can reduce accuracy, making the same-farm validation result an unreliable estimate of performance across customers' farms. [1 mark]

Evaluate on labelled photographs from separate farms and seasons, chosen to represent the intended use. [1 mark] Broaden the training set to include those conditions, or adapt the model using a labelled subset from a new farm while retaining separate examples for testing. [1 mark]

Question 2

(a) Knowledge, Comprehension

Here x is a real creature image and z is a random vector drawn from a simple distribution such as a Gaussian. [1 mark] The generator turns z into an image $G(z)$, while $D(x)$ estimates the probability that its input is real. [1 mark]

Maximising $\log D(x)$ rewards the discriminator for recognising real images. [1 mark] Maximising $\log(1 - D(G(z)))$ rewards it for rejecting generated images, so the two terms together train it to distinguish the two sources. [1 mark] The generator minimises the second term, encouraging images that receive a high discriminator probability. The first term is independent of its weights. [1 mark]

In each iteration, sample real images and fresh noise, then update the discriminator using real and detached generated images. [1 mark] Next evaluate the updated discriminator on the generated images and backpropagate through it to update the generator, keeping discriminator weights fixed during that update. [1 mark]

The practical uses discriminator logits and the non-saturating generator loss $-\log D(G(z))$. Describing this variant is also acceptable.

(b) Knowledge, Comprehension, Application

Diffusion. Sample a creature image and a random time, then corrupt the image with Gaussian noise. [1 mark] A fixed schedule determines the corruption strength, with the last time corresponding to approximately pure noise. [1 mark] Train a network that receives the noisy image and time to predict the added noise using mean squared error. [1 mark]

To generate, start from Gaussian noise. [1 mark] Work backwards through the schedule, repeatedly predicting the noise and using it in the reverse-process update. [1 mark] In the DDPM sampler used here, add fresh Gaussian noise at each reverse step except the last. [1 mark]

Flow matching: an alternative answer. Pair a real image x_1 with independent Gaussian noise x_0 , and sample $t \in [0, 1]$. [1 mark] Form the intermediate state $x_t = (1 - t)x_0 + tx_1$. [1 mark] Train a network receiving x_t and t to predict $x_1 - x_0$ using mean squared error. [1 mark]

Generation starts with Gaussian noise at $t = 0$. [1 mark] Numerically integrate the learned velocity field until $t = 1$, for example with Euler steps. [1 mark] Each update adds the current predicted velocity multiplied by the step length to the evolving image. [1 mark]

(c) Analysis, Evaluation

A well-trained GAN can generate a sharp image in one forward pass, making sampling fast. [1 mark] However, it can collapse onto a small range of outputs, as illustrated by the repeated blue bird, and consequently miss much of the training distribution. [1 mark]

A VAE also samples in one decoder pass. Its likelihood-based objective encourages reconstruction across the dataset, although coverage is not guaranteed. [1 mark] A basic VAE with a pixelwise Gaussian likelihood often averages uncertain fine detail and produces blur, particularly when the latent representation is strongly constrained. [1 mark]

Diffusion and flow matching can combine detailed images with broad coverage when the model and training are adequate. [1 mark] Their usual iterative samplers require repeated network evaluations, making them slower than a similarly sized single-pass generator. [1 mark]

For the artists' tool, diffusion or flow matching is a suitable choice because the available time permits iterative sampling and both quality and variety matter. [1 mark] For the phone, a small GAN is a plausible choice if sharpness is the priority, provided actual latency and diversity are checked. A small VAE is defensible if its reconstruction quality is sufficient for the game. [1 mark]

These are common tradeoffs, not guarantees of a model family's behaviour. Other recommendations earn credit when justified in terms of quality, coverage and measured sampling cost.

(d) Application, Synthesis, Evaluation

Label the training images by creature type and provide the label alongside the noisy image and time, using a one-hot vector or learned embedding. [1 mark] During generation, supply the requested type at every step. [1 mark]

To blend types, interpolate their condition vectors, for example $(1 - s)e_{\text{bird}} + se_{\text{fish}}$. Reuse the same initial noise for each value of s , including the same later noise draws in a stochastic diffusion sampler, to isolate the condition's effect. [1 mark] Training on pure types does not specify what intermediate embeddings should mean, so blends may be plausible, ambiguous or malformed and require inspection. [1 mark]

Question 3**(a) Knowledge, Application**

Encode each character as an integer selecting a learned embedding. [1 mark] At each position, use the next character as the target. [1 mark] For hello, this gives inputs [h,e,l,l,o] and targets [e,l,l,o], providing four predictions from the word. [1 mark] Train the resulting character logits with cross-entropy, which averages the negative log-probability assigned to each correct next character. [1 mark]

(b) Comprehension, Analysis

Combine each token embedding with position information, then apply learned linear maps to obtain queries, keys and values. [1 mark] At a position, take dot products between its query and the keys at available positions. [1 mark] Scale these scores by $1/\sqrt{d}$ and apply softmax to obtain weights summing to one. [1 mark] Their weighted sum of the value vectors combines information from the earlier characters. [1 mark] Learning the projections lets the model adapt which information is selected and how it is represented. [1 mark]

During training, the target character is present at the next position in the input passage. A causal mask gives future positions zero weight, preventing the network from using that target directly. [1 mark] This matches generation, when future characters are unavailable, and avoids training the model to rely on information it will lack in use. [1 mark]

(c) Analysis, Evaluation

Adjacent windows share 255 of their 256 characters, so randomly assigned validation windows are likely to overlap almost completely with training windows. [1 mark] The resulting loss measures predictions on text substantially exposed during training, making it an optimistic estimate for new messages. It does not by itself prove that every message was memorised. [1 mark]

Split the collection into disjoint messages before constructing windows. [1 mark] Holding out entire volunteers is preferable when the intended test concerns new users, because it also separates their personal wording and habits. [1 mark]

(d) Comprehension, Application

Give the model the typed prefix and convert the final position's logits to next-character probabilities. [1 mark] Sample a character, append it, and repeat with the extended prefix until the suggestion is long enough, retaining only as much context as the model accepts. [1 mark]

Divide logits by temperature before softmax. Below one, this concentrates probability on likely continuations, which can become repetitive. [1 mark] Above one, it spreads probability more broadly, tending to increase variety alongside spelling and grammatical errors. [1 mark] Greedy selection is also acceptable in place of sampling.

(e) Analysis, Synthesis, Evaluation

Model A's similar but relatively high training and validation losses suggest underfitting, which could reflect insufficient capacity or incomplete optimisation. [1 mark] Model C fits training text much more closely but has worse validation loss than B, so its larger gap is evidence of overfitting rather than better prediction on new messages. [1 mark] Its 6,509,312 parameters also exceed the storage limit. [1 mark]

Recommend B: its 890,368 parameters fit the allowance and its validation NLL of 1.45 is the lowest of the eligible models. [1 mark] Dropout or weight decay could reduce overfitting, with the strength chosen using validation loss. [1 mark] Sharing the input embedding and output projection weights could free space for additional capacity within the budget. [1 mark]

More representative training data, a decaying learning rate or early stopping are also suitable changes when their purpose is explained. Improvements must be checked rather than assumed.

Question 4**(a) Knowledge, Comprehension**

Use a network whose input is a pixel's two coordinates, conveniently scaled to $[-1, 1]$. [1 mark] Its three outputs predict RGB values on the image's $[0, 1]$ intensity scale. [1 mark] The photograph itself supplies 65,536 coordinate-colour training pairs. [1 mark]

At each update, sample a batch of coordinates and predict their colours. [1 mark] Backpropagate the mean squared colour error and update the weights with an optimiser such as Adam. [1 mark] To reconstruct the photograph later, evaluate the stored network at every pixel coordinate and arrange its outputs in the original grid, without reading the original pixel values. [1 mark]

(b) Application

Including biases, the first layer contains $2 \times 64 + 64 = 192$ parameters. [1 mark] The second contains $64 \times 64 + 64 = 4,160$. [1 mark] Adding the output layer's $64 \times 3 + 3 = 195$ gives 4,547 in total. [1 mark]

Direct RGB storage requires $256 \times 256 \times 3 = 196,608$ values, about 43 times as many. Precision matters when comparing storage: 4,547 float32 weights occupy 18,188 bytes, compared with 196,608 bytes for uncompressed eight-bit RGB pixels, before metadata. [1 mark]

(c) Analysis, Synthesis

A ReLU coordinate network represents a piecewise-affine function, so sharp boundaries and fine texture require appropriate changes between its affine regions. [1 mark] Small coordinate networks often fit smooth, broad structure more readily than rapidly varying detail. Blur can reflect both limited capacity and optimisation, and the observation alone does not distinguish them. [1 mark]

Replacing ReLU with sine activations adds no parameters. [1 mark] With suitable frequency scales and initialisation, those units can represent rapid coordinate variation more readily, as in SIREN. [1 mark]

Alternatively, allocate part of the existing budget to multiresolution feature grids and reduce the decoder accordingly. [1 mark] Interpolated fine-grid features can describe local detail, while bounded hash tables limit storage at the cost of collisions. Every stored table value must count towards the unchanged budget. [1 mark]

Complex Gabor activations are another acceptable change if both real and imaginary parameters are counted.

(d) Application

With MSE 0.01, PSNR is $-10 \log_{10}(0.01) = 20$ dB. [1 mark] Reaching 30 dB requires MSE at most $10^{-3} = 0.001$, a tenfold reduction from the stated error. [1 mark]

(e) Analysis, Evaluation

A coordinate network can be evaluated on a 1024×1024 grid because it defines a colour at each coordinate. [1 mark] However, the additional coordinates provide no new observations. Many detailed images could agree with the original low-resolution data, so added texture may be an artefact of the fitted function rather than recovered detail. [1 mark]

Test the claim using high-resolution photographs kept as reference targets. Fit only their downsampled versions, applying the same downsampling operation to predictions in the loss, then compare the reconstructed high-resolution images with both the reference and bicubic upsampling. [1 mark]

(f) Analysis, Synthesis

Each set of weights is fitted to one photograph independently. [1 mark] Consequently, processing millions of photographs repeats the optimisation millions of times instead of reusing a learned encoding procedure. [1 mark]

Train a shared autoencoder on many aerial photographs, with an encoder that maps each photograph to a compact code and a decoder that reconstructs it. [1 mark] A new photograph then needs one encoder pass, and only its code is stored per image. The shared decoder weights must also be retained, with their cost spread across the collection. Reconstruction quality depends on code size and how well the training data represent future photographs. [1 mark]

A shared coordinate decoder conditioned on per-image codes is also acceptable. If those codes are optimised separately, compression still requires fitting each new code unless an encoder is provided.